

Diagram illustrating the decomposition of a 4x4x4 tensor into a sum of 8 rank-1 tensors.

The tensor is represented by a 4x4x4 cube labeled $\langle 2, 2, 2 \rangle$ and $\in \mathbb{R}^{4 \times 4 \times 4}$.

The decomposition is shown as a sum of 8 rank-1 tensors, arranged in two rows of four:

- Row 1:
 - Tensor 1: Blue vertical bar labeled a_{11} , orange horizontal bar labeled b_{11} , green diagonal bar labeled c_{11} .
 - Tensor 2: Blue vertical bar labeled a_{12} , orange horizontal bar labeled b_{21} , green diagonal bar labeled c_{11} .
 - Tensor 3: Blue vertical bar labeled a_{21} , orange horizontal bar labeled b_{11} , green diagonal bar labeled c_{21} .
 - Tensor 4: Blue vertical bar labeled a_{22} , orange horizontal bar labeled b_{21} , green diagonal bar labeled c_{21} .
- Row 2:
 - Tensor 5: Blue vertical bar labeled a_{11} , orange horizontal bar labeled b_{12} , green diagonal bar labeled c_{12} .
 - Tensor 6: Blue vertical bar labeled a_{12} , orange horizontal bar labeled b_{22} , green diagonal bar labeled c_{12} .
 - Tensor 7: Blue vertical bar labeled a_{21} , orange horizontal bar labeled b_{12} , green diagonal bar labeled c_{22} .
 - Tensor 8: Blue vertical bar labeled a_{22} , orange horizontal bar labeled b_{22} , green diagonal bar labeled c_{22} .

The decomposition is represented by the equation:

$$\langle 2, 2, 2 \rangle \in \mathbb{R}^{4 \times 4 \times 4} = \text{Tensor 1} + \text{Tensor 2} + \text{Tensor 3} + \text{Tensor 4} + \text{Tensor 5} + \text{Tensor 6} + \text{Tensor 7} + \text{Tensor 8}$$